Investigating Under-Reliance on Automation

Lara Suraci
with Robin Cubitt and Chris Starmer

31 March 2022

Motivation

"For automation to be beneficial and successfully integrated into society it must first be used and accepted by humans. We can develop as much as we want, what good does it do if nobody uses it?"

– Liehner et al. (2021)

Previous Research

- **Suboptimal reliance** on automated agents is a well-documented phenomenon (Dzindolet et al. 2002; Bonnefon et al. 2016; Ben-David & Sade, 2020)
- So far, this has been explored mainly in the context of **trust in automation**:

  (1) in the Human Factors literature (Dzindolet et al. 2003; Lee & See, 2004; Hoff & Bashir, 2015; Lee & Rich, 2021)

  (2) in Economics and Psychology (Gogoll & Uhl, 2018; Schniter et al. 2020)

CASA – Computers Are Social Actors

- Nass & Moon's (2000) **CASA paradigm** (Computers Are Social Actors) states that "individuals mindlessly apply social rules and expectations to computers" in the form of:

  (1) human social categories

  (2) overlearned social behaviours

  (3) premature cognitive commitments

CASA – Computers Are Social Actors

- Nass & Moon's (2000) **CASA paradigm** (Computers Are Social Actors) states that "individuals mindlessly apply social rules and expectations to computers" in the form of:

  (1) human social categories

  (2) **overlearned social behaviours**

  (3) premature cognitive commitments

- For example: evidence of **reciprocal behaviour** towards automated agents (Fogg & Nass, 1997; Lammer et al. 2014; Lee & Liang, 2015; Sandoval et al. 2016; Kim et al. 2021)

  → Why might a decision-maker reject assistance from a **"social actor"**?

Beliefs and Agency

- Cameron et al. (2015) and Liehner et al. (2021) highlight the importance of context for investigations into automation reliance, including the role of **beliefs** about the capabilities of the automated agent

  → Possible **misalignment** between the assistance offered to the decision-maker and their beliefs about the actual capabilities of the automated agent

Beliefs and Agency

- While human intentionality is intrinsic, that of automated agents is "derived" from the **intentionality of the programmer** (Searle, 1980; 1983; 1992)
- Cf. betrayal aversion towards **inanimate objects** (Koehler & Gershoff, 2003) which Cubitt et al. (2017) hypothesise "reflects a feeling of being let down by the designer or supplier of the object"
- Johnson (2006) argues that automated agents should thus not be dismissed from the **realm of morality** and Gray et al. (2007) finds that subjects do ascribe a moderate degree of **agency** to the robot in their study

  → Similar to its human counterpart, an offer of automated assistance could be perceived as **kind** or **unkind**

Research Questions

(1) How does a **misalignment** between the offered level of automated assistance and an individual's beliefs about the capabilities of the automated agent affect automation reliance?

(2) What role does the degree of **agency** ascribed to the automated agent in choosing the offered level of automated assistance play?

Under-Reliance on Automation

- Non-reliance of a human decision-maker (D) on a computer (C) can be classed as **under-reliance** if:

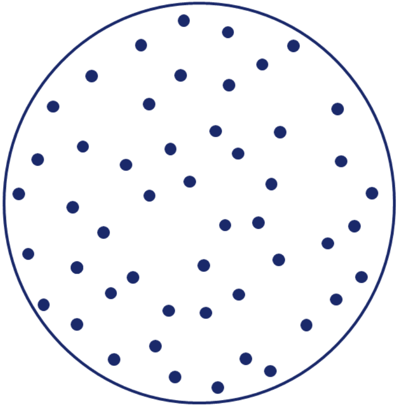
$$E(\text{D's Payoff} \mid \text{Reliance on C}) > E(\text{D's Payoff} \mid \text{Non-Reliance on C})$$

- Let q be the success rate of D and p that of C in a simple experimental task, such that if $p > q$ non-reliance **always** constitutes under-reliance
- Setting $q = 0.5$ (and $p > 0.5$) should render the **inferiority** of D's abilities relative to C salient regardless of their confidence in their own abilities

  $\rightarrow$ **perceptual task** from Hollard et al. (2016)

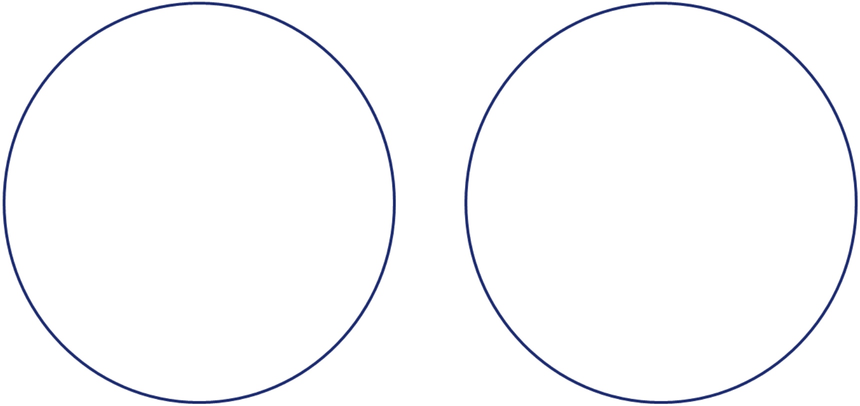
Perceptual Task

- Which of these two circles contains the **higher number** of dots?



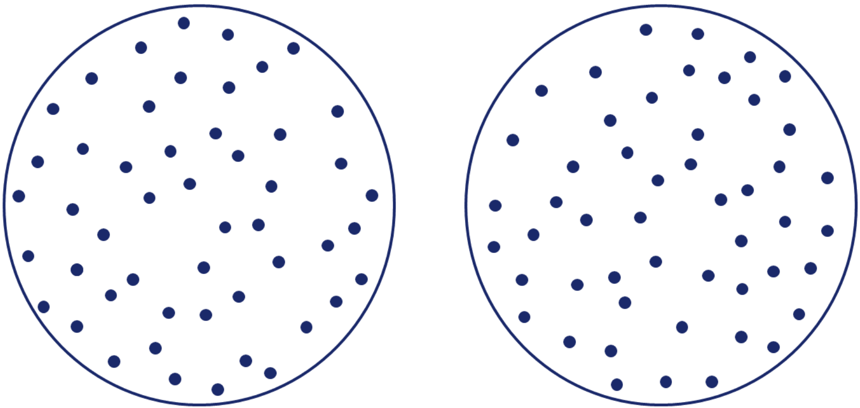
Perceptual Task

- Which of these two circles contains the **higher number** of dots?



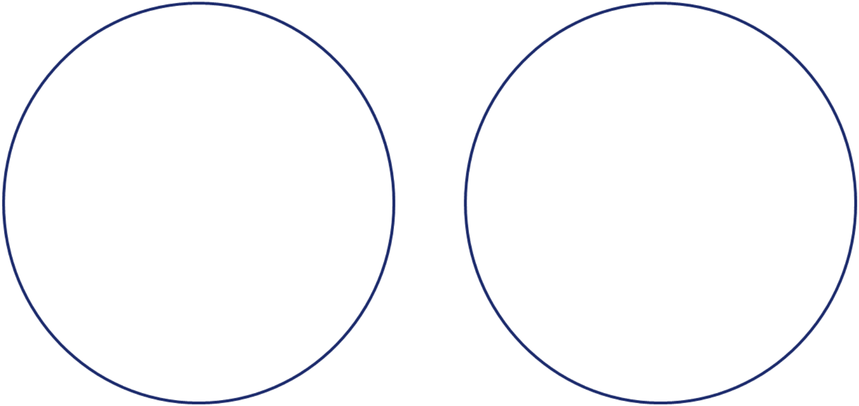
Perceptual Task

- Which of these two circles contains the **higher number** of dots?



Perceptual Task

- Which of these two circles contains the **higher number** of dots?



Experimental Design

- **Stage 1 –** D performs a series of ten practise rounds of the perceptual task

  Stage 2 – D learns p and then decides to have their payoff in Stage 3 depend on their own performance or that of C

  Stage 3 – D and C perform another ten rounds of the perceptual task

- In Stage 2, in addition to the realised p, D may also learn about a set of **alternative success rates** that C could have performed at

Treatments

- Treatments vary:
  (1) where the selected p ranks relative to the alternative success rates
  (2) the (perceived) agency of the computer in selecting p
- Using a **non-uniform random device** to select p from the set of success rates, we will make sure that $p = 0.6$ in most cases

	P = 0.6 BAD OUTLIER	P = 0.6 GOOD OUTLIER
P IS NOT CHOSEN FROM A SET	T1	
P IS CHOSEN FROM A SET BY THE COMPUTER	T2	T3
P IS CHOSEN FROM A SET BY THE EXPERIMENTER	T4	T5

Treatments

- **T1:** $p = \{\mathbf{0.60}\}$

  $\rightarrow$ "The computer selects the correct solution 60% of the time."

- **T2 & T4:** $p = \{\mathbf{0.60}, 0.67, 0.68, 0.69, 0.70\}$

- **T3 & T5:** $p = \{0.50, 0.51, 0.52, 0.53, \mathbf{0.60}\}$

  $\rightarrow$ "The computer has chosen to select the correct solution 60% of the time."

  $\rightarrow$ "The computer has been programmed by the experimenter to select the correct solution 60% of the time."

Additional Measurements

- In between Stages 2 and 3, we include a survey which measures:

  (1) expectations of D's own performance and that of C

  (2) social and non-social emotions based on the Positive and Negative Affect Schedule (Watson et al. 1988) about the offer of automated assistance

  (3) the perceived **kindness** of the offer of automated assistance

  (4) measurement of **risk and ambiguity preferences** and **confidence**

Hypotheses

H1. We observe some under-reliance even when p is not chosen from a set.

H2a. Under-reliance is higher when p is a bad outlier than when p is a good outlier.

H2b. Under-reliance is higher when p was chosen by the experimenter if p is a bad outlier and lower if p is a good outlier.

H3a. Social emotions are stronger when p was chosen by the experimenter than when p was chosen by the computer.

H3b. Perceived (un)kindness of the offer is more pronounced when p was chosen by the experimenter than when p was chosen by the computer.

H4. Positive (negative) social and non-social emotions are positively (negatively) correlated with automation reliance.

H5. Perceived (un)kindness of the offer is (negatively) positively correlated with automation reliance.

Summing Up

- Suboptimal reliance on automated agents could negatively impact the benefits of future advances in the fields of automation and AI
- Nass and Moon's (2000) CASA paradigm suggests that humans often treat computers as social actors and mindlessly apply overlearned social behaviours
- Given the formation of beliefs about the true capabilities of automated agents and the possibility of humans ascribing them some degree of agency, an offer of automated assistance might be considered kind or unkind
- We therefore propose an experiment in which we manipulate:

  (1) the framing of an offer of automated assistance via different sets of non-realised alternative offers

  (2) the degree of agency of the automated agent in making this offer

Thank You!